

Title - **British Airways Customer Review Analysis**

Introduction:

British Airways is one of the leading international airlines, serving millions of passengers every year across different countries and destinations. Every passenger shares valuable feedback based on their travel experience, including cabin service, seat comfort, food quality, entertainment, and overall satisfaction. These customer reviews help the airline understand passenger expectations and identify areas where improvements are needed.

However, customer review data is usually large and difficult to analyze manually. Looking at thousands of reviews individually is time-consuming and does not easily reveal meaningful trends or patterns. Therefore, businesses use Business Intelligence (BI) tools like Power BI to convert raw data into interactive dashboards that support quick analysis and better decision-making.

In this project, Power BI was used to analyze British Airways customer review data. The dataset was cleaned and transformed before creating interactive dashboards to visualize customer satisfaction, service quality, traveller preferences, aircraft review distribution, and value-for-money ratings. Different Power BI visuals such as KPI cards, donut charts, ribbon charts, decomposition trees, treemaps, pie charts, and bar charts were used to present the information clearly.

The main purpose of this project is to identify customer satisfaction trends, understand passenger behaviour, and provide business recommendations that can help British Airways improve the overall travel experience.

Business Objective:

The primary objective of this project is to analyze customer reviews of British Airways and identify the major factors that influence passenger satisfaction. Every day, thousands of passengers share their travel experiences by rating different airline services such as seat comfort, cabin staff, food quality, entertainment, ground service, and value for money. This feedback contains useful information that can help the airline improve its services and increase customer satisfaction.

To achieve this objective, the customer review dataset was first cleaned and preprocessed using **Python in Jupyter Notebook**. During preprocessing, the data was checked for missing values, duplicate records, incorrect data types, and inconsistent values to ensure that the dataset was accurate and suitable for analysis. After cleaning, the processed data was imported into **Power BI**, where relationships between the datasets were created and DAX measures were developed to calculate important business metrics.

Interactive dashboards were then created to visualize customer satisfaction, service quality, traveller preferences, aircraft performance, and value for money. These dashboards allow users to understand customer behaviour, compare different passenger categories, identify strengths and weaknesses in airline services, and support data-driven business decisions.

The overall goal of this project is to convert raw customer feedback into meaningful business insights that help British Airways improve passenger experience, enhance service quality, and maintain high customer satisfaction levels.

Business Problem Statement:

British Airways receives customer reviews from passengers travelling across different countries, aircraft, seat classes, and traveller categories. Since the number of reviews is very large, manually analyzing this information becomes difficult and time-consuming.

As a result, management may find it challenging to identify customer expectations, understand service quality, or recognize the main reasons behind passenger satisfaction or dissatisfaction.

The airline management wants answers to several important business questions. They need to know which seat type provides the best customer experience, which traveller category is the most satisfied, whether food quality and cabin staff service influence overall ratings, which aircraft receive the highest number of reviews, and whether passengers believe they receive good value for the money they spend.

They also want to identify areas where improvements are required so that future customer satisfaction can be increased.

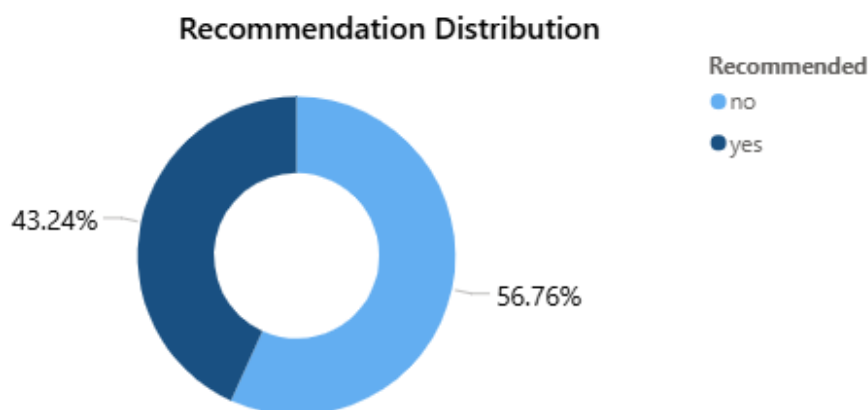
To solve these problems, this project uses **Python** for data cleaning and preprocessing and **Power BI** for interactive data visualization. Python helps prepare a clean and reliable dataset by handling missing values, removing duplicate records, and correcting data formats.

The cleaned dataset is then imported into Power BI, where interactive dashboards are created using DAX measures, charts, and slicers.

These dashboards help management quickly understand customer feedback, compare different services, and make informed business decisions based on data rather than assumptions.

Dashboard Analysis:

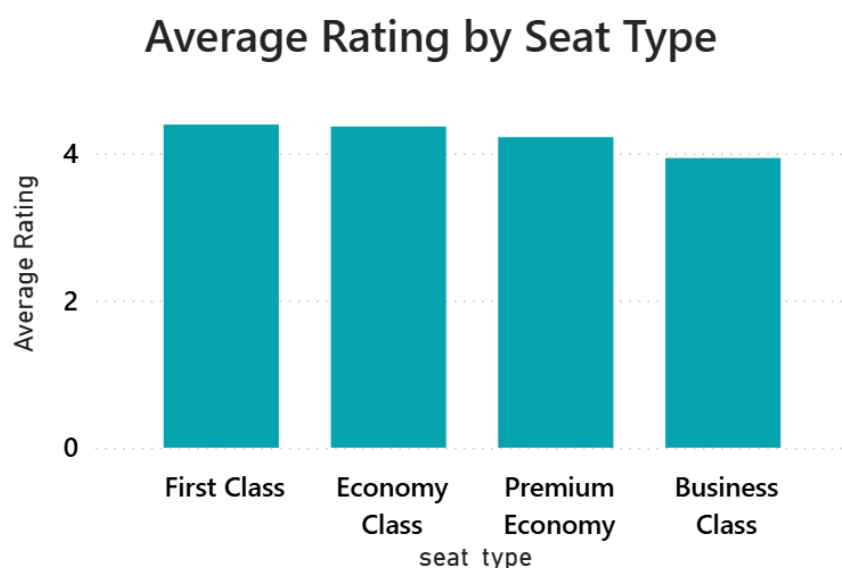
Chart 1 – Recommendation Distribution



Insight:

The Recommendation Distribution chart shows whether customers would recommend British Airways to others. Around **43% of passengers recommended** the airline, while the remaining customers did not. This indicates that customer opinions are mixed. Although many passengers are satisfied with their travel experience, there is still room for improvement. British Airways should focus on enhancing service quality, customer support, and overall travel experience to increase the recommendation rate and improve customer loyalty.

Chart 2 – Average Rating by Seat Type

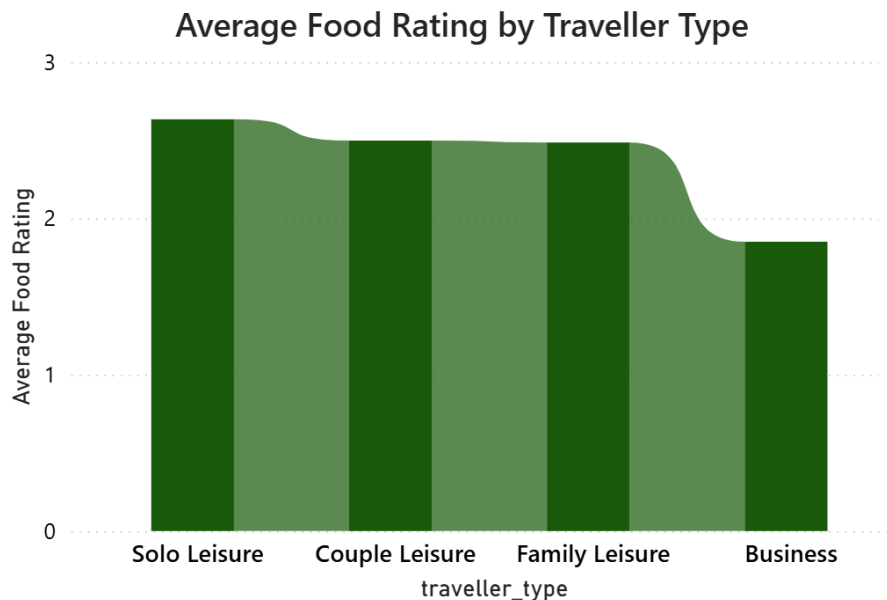


Insight:

This chart compares the average customer ratings across different seat types. **First Class has the highest average rating**, while **Business Class has the lowest** among the four seat

categories. This shows that premium services are generally appreciated by passengers. British Airways can improve customer satisfaction by enhancing the Business Class experience and maintaining the quality of services offered in First Class.

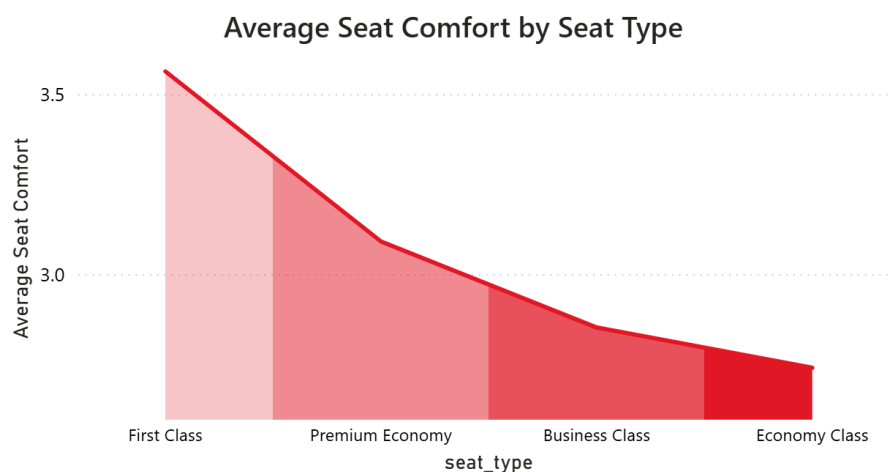
Chart 3 – Food Rating by Traveller Type



Insight:

The chart shows food ratings given by different traveller types. **Solo Leisure travellers provide the highest food ratings**, while **Business travellers give comparatively lower ratings**. This suggests that customer expectations vary across traveller categories. Improving meal quality, menu options, and consistency can help increase customer satisfaction, especially among frequent business travellers.

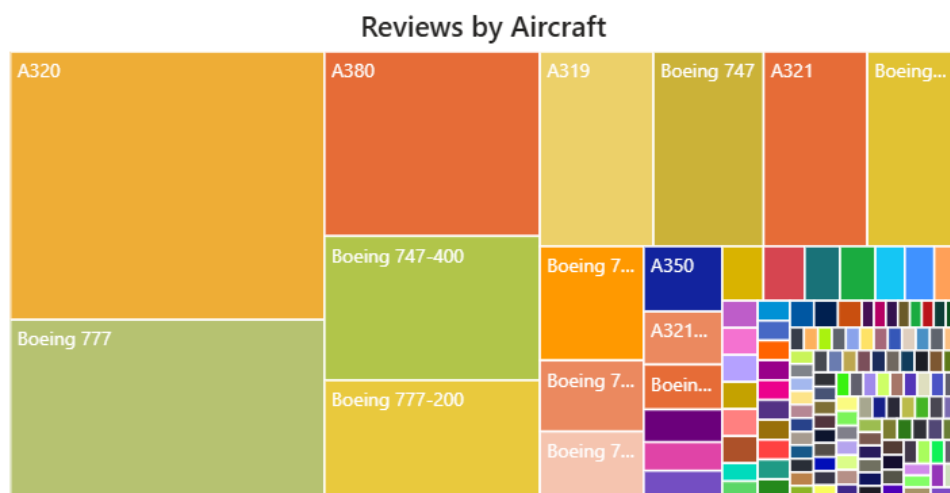
Chart 4 – Average Seat Comfort by Seat Type



Insight:

Seat comfort ratings are highest for **First Class** and gradually decrease towards **Economy Class**. This indicates that passengers value comfort as an important part of their travel experience. Improving seating arrangements, legroom, and overall comfort in Economy and Business Class could help improve customer satisfaction and encourage repeat travel.

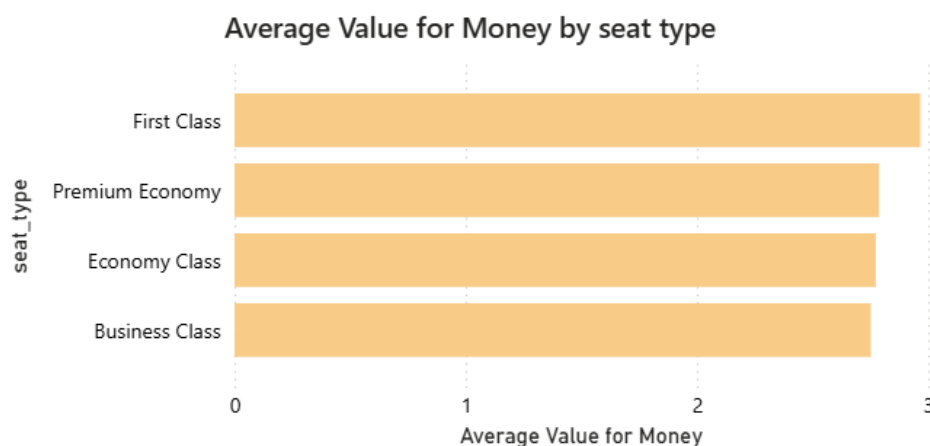
Chart 5 – Reviews by Aircraft



Insight:

The treemap shows the number of customer reviews received by different aircraft. The **A320** and **Boeing 777** have the highest number of reviews, indicating that these aircraft are used more frequently or serve more passengers. Monitoring customer feedback for these aircraft can help British Airways identify common issues and improve the travel experience for a larger number of customers.

Chart 6 –Average Value for Money by Seat Type



Insight:

This chart compares passengers' opinions about value for money across different seat types. **First Class receives the highest value-for-money rating**, while **Business and Economy Class receive slightly lower ratings**. This suggests that passengers in premium classes are more satisfied with the services they receive. Improving service quality in Business and Economy Class can help increase customer satisfaction and provide better value to passengers.

Business Recommendations:

Based on the dashboard analysis, the following recommendations can help British Airways improve customer satisfaction and service quality:

- Improve the overall experience in **Business Class**, as it received comparatively lower customer ratings than expected.
- Focus on improving **seat comfort**, especially in Economy and Business Class, since comfort has a direct impact on customer satisfaction.
- Enhance the quality and variety of **onboard food**, particularly for Business travellers who have higher service expectations.
- Continue maintaining the high service standards provided in **First Class**, as it consistently receives better customer ratings.
- Regularly monitor customer feedback from frequently used aircraft such as the **A320** and **Boeing 777**, as improvements on these aircraft will benefit a larger number of passengers.
- Use customer review data to identify service issues early and make data-driven decisions for continuous improvement.
- Continue using interactive Power BI dashboards to monitor customer satisfaction and support management in making faster and more effective business decisions.

Conclusion:

This project successfully demonstrates how customer review data can be transformed into meaningful business insights using **Python and Power BI**. Python was used in **Jupyter Notebook** to clean and preprocess the raw datasets by handling missing values, checking data quality, removing duplicate records, and preparing the data for analysis.

The cleaned data was then imported into **Power BI**, where DAX measures, relationships, and interactive dashboards were created to analyze customer satisfaction, service quality, traveller preferences, aircraft reviews, and value for money.

The dashboards provide a clear understanding of customer opinions and help identify both strengths and areas requiring improvement. The analysis shows that factors such as seat comfort, food quality, cabin staff service, and value for money have a significant influence on customer satisfaction.

Interactive visualizations make it easier to compare different traveller types, seat classes, and aircraft, allowing management to identify important trends quickly. The recommendations generated from this analysis can help British Airways improve passenger experience, enhance operational performance, and make better data-driven decisions.

This project also demonstrates the complete data analytics workflow, starting from data preprocessing in Python to dashboard development in Power BI. It highlights the importance of converting raw data into useful business insights that support continuous service improvement. Overall, this project strengthened my understanding of data cleaning, data visualization, business intelligence, and analytical thinking while solving a real-world business problem.